

Self-Study Assignment 2

Canonical LR(1) / CLR(1) Parser Construction

Course Code: CSE 420

Course Title: Compiler Design

Section: 18

Assignment Topic: Study the construction of the Canonical LR(1), i.e. CLR(1), parser from the reference textbook (Section 4.7.1 to Section 4.7.3) and show the full construction process for the assigned grammar.

Instructions:

1. This is a self-study assignment on **CLR(1) parser construction**.
2. Each student must solve **only one** assigned problem.
3. Let the last two digits of the student's ID be N .
4. Compute:

$$k = N \bmod 20$$

5. The assigned problem will be selected using the following rule:

k value	Assigned Problem
0	Problem 1
1	Problem 2
2	Problem 3
3	Problem 4
4	Problem 5
5	Problem 6
6	Problem 7
7	Problem 8
8	Problem 9
9	Problem 10
10	Problem 11
11	Problem 12
12	Problem 13
13	Problem 14
14	Problem 15
15	Problem 16
16	Problem 17
17	Problem 18
18	Problem 19
19	Problem 20

What to Do

For the assigned grammar, show the complete CLR(1) parser construction process:

- Augment the grammar.
- Write the necessary **FIRST** sets needed for LR(1) lookaheads.
- Construct the initial LR(1) item set I_0 using **CLOSURE**.
- Construct the remaining LR(1) item sets using **GOTO**.
- Draw the CLR(1) automaton / transition diagram.
- Construct the **ACTION** and **GOTO** parsing table.
- State whether the grammar is CLR(1) without conflict.

Problems

Problem 1

Construct the CLR(1) parser for:

$$S \rightarrow bA \quad A \rightarrow cA \mid d$$

Problem 2

Construct the CLR(1) parser for:

$$S \rightarrow bA \quad A \rightarrow c \mid cA$$

Problem 3

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow x \quad B \rightarrow y$$

Problem 4

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow xA \mid x \quad B \rightarrow y$$

Problem 5

Construct the CLR(1) parser for:

$$S \rightarrow Az \quad A \rightarrow bA \mid c$$

Problem 6

Construct the CLR(1) parser for:

$$S \rightarrow pB \quad B \rightarrow qB \mid r$$

Problem 7

Construct the CLR(1) parser for:

$$S \rightarrow DD \quad D \rightarrow mD \mid n$$

Problem 8

Construct the CLR(1) parser for:

$$S \rightarrow pS \mid q$$

Problem 9

Construct the CLR(1) parser for:

$$S \rightarrow At \quad A \rightarrow rA \mid s$$

Problem 10

Construct the CLR(1) parser for:

$$S \rightarrow A \quad A \rightarrow pA \mid qA \mid r$$

Problem 11

Construct the CLR(1) parser for:

$$S \rightarrow pSq \mid r$$

Problem 12

Construct the CLR(1) parser for:

$$S \rightarrow uA \quad A \rightarrow vA \mid w$$

Problem 13

Construct the CLR(1) parser for:

$$S \rightarrow pB \mid qA \quad A \rightarrow r \quad B \rightarrow s$$

Problem 14

Construct the CLR(1) parser for:

$$S \rightarrow Ap \mid Bq \quad A \rightarrow r \quad B \rightarrow r$$

Problem 15

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow m \mid mm \quad B \rightarrow n$$

Problem 16

Construct the CLR(1) parser for:

$$S \rightarrow pA \mid qB \quad A \rightarrow r \quad B \rightarrow s$$

Problem 17

Construct the CLR(1) parser for:

$$S \rightarrow A \quad A \rightarrow pB \quad B \rightarrow qC \quad C \rightarrow r$$

Problem 18

Construct the CLR(1) parser for:

$$S \rightarrow pA \quad A \rightarrow qC \mid r \quad C \rightarrow s$$

Problem 19

Construct the CLR(1) parser for:

$$S \rightarrow pA \mid pB \quad A \rightarrow r \quad B \rightarrow s$$

Problem 20

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow p \mid q \quad B \rightarrow r \mid s$$

Submission

1. The assignment must be submitted in **handwritten** form.
2. Write all LR(1) items clearly with lookaheads.
3. Show the CLOSURE and GOTO steps in an organized way.
4. Draw the automaton / state transition diagram neatly.
5. Prepare the final ACTION and GOTO table clearly.